

# GymTrack

## Gym Membership Management System

|                   |                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------|
| Project Title     | GymTrack — Gym Membership Management System                                               |
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| Date              | July 15, 2026                                                                             |
| GitHub Repository | <a href="https://github.com/louisflim/gymtrack">https://github.com/louisflim/gymtrack</a> |

Repository link: <https://github.com/louisflim/gymtrack>

## 1. Development Progress Summary

GymTrack is a full-stack gym membership system with a Spring Boot backend, React web client, and Android mobile app. Early work established authentication, dashboards, plans, members, memberships, PayMongo payments, and QR attendance. Mid-phase work reorganized the codebase into vertical feature slices (auth, plans, members, membership, payments, attendance, staff, dashboard) shared across web and mobile. Recent work focused on account security, safer admin member removal, production-ready API hardening, clearer errors, and a shared Settings experience for all roles.

Progress milestones:

- Backend + web + mobile foundations (auth, gym model, dashboards).
- Feature vertical slices for registration, login, staff creation, plans, members, membership, payments, attendance, and admin KPIs.
- PayMongo checkout integration (mock/test-friendly for school deployment).
- QR enrollment and attendance check-in/check-out for staff and members.
- Staff forced password change on first login; reusable change-password API.
- API method security and production secret/profile guards.
- Admin “remove member from gym” (unenroll) instead of hard-deleting the user account.
- Settings menu (web + mobile): change password and self-delete account.

## 2. Newly Implemented Features

### 2.1 Account Settings (Web & Mobile)

A Settings tab is available to Admin, Staff, and Member. Users can change their password and permanently delete their own account (password confirmation + confirm dialog). Light/dark theme toggles were

intentionally not included. Shared components keep web and mobile behavior aligned (ChangePasswordForm / SettingsPanel on web; layout\_settings\_panel + SettingsViewModel on mobile).

## 2.2 Staff First-Login Password Reset

When an admin creates a staff account, mustChangePassword is set. Staff must set a new password via POST /api/auth/change-password before using the dashboard. Web and mobile redirect accordingly.

## 2.3 Soft Remove Member from Gym

Gym owners remove members from their gym without deleting the users row. Memberships are expired, gym\_id is cleared, and the login account remains so the person can join again later. Confirm dialogs on web and mobile clarify this behavior.

## 2.4 Security Hardening for Deployment

Added @EnableMethodSecurity, @PreAuthorize on controllers, HTTP role matchers, production profile validation for secrets, CORS/webhook guards, and safer PayMongo mock ownership checks.

## 2.5 UX Reliability Fixes

Fixed login/register buttons that stayed disabled until re-render, staff login after password change, and friendlier API error messages across clients.

# 3. How Features Were Integrated

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Integration follows a shared API contract: React (axios) and Android (Retrofit) call the same Spring Boot endpoints under /api/\*. Session tokens (JWT) are stored in localStorage (web) and DataStore (mobile). Feature modules are organized as vertical slices so UI, repositories, and backend packages stay aligned (e.g., feature/auth, feature/settings, feature/members).

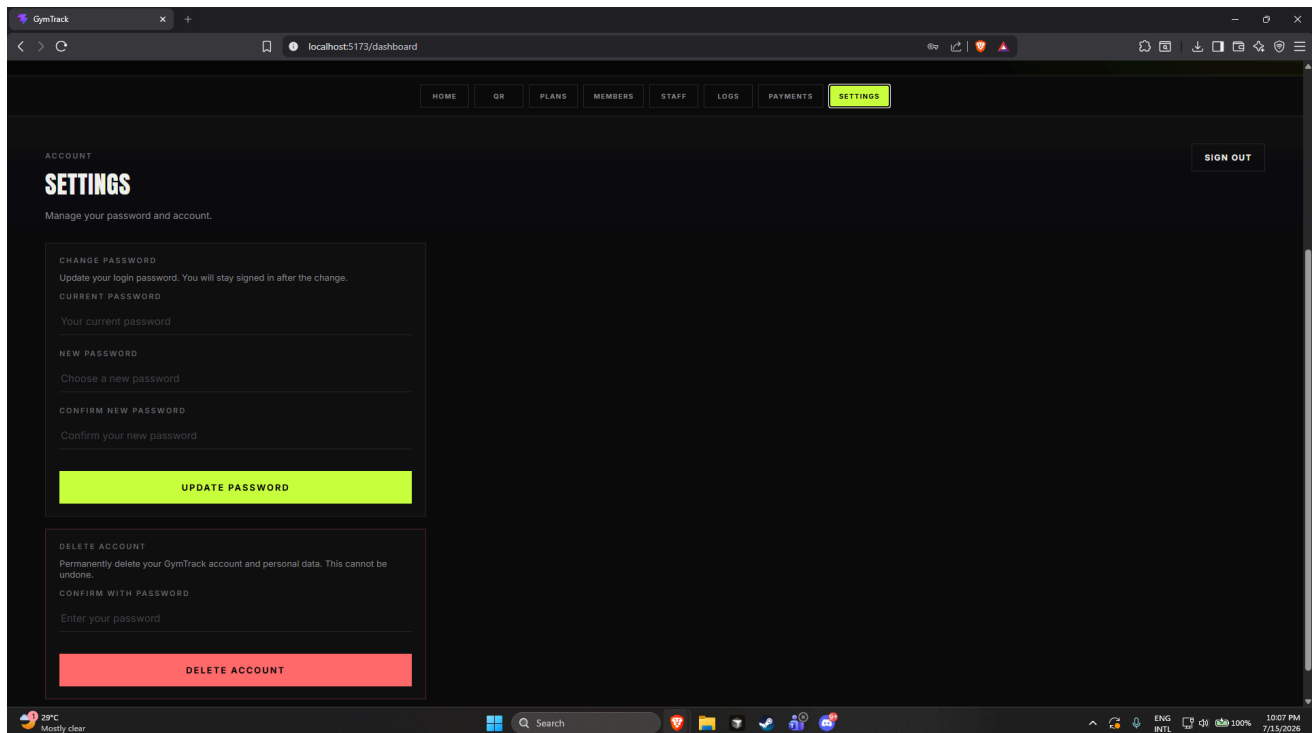
- Settings → POST /api/auth/change-password (existing) and DELETE /api/auth/account (new AccountService).
- Admin member remove → DELETE /api/members/{id} now unenrolls (MemberService) instead of hard delete.
- Staff password gate → AuthResponse.mustChangePassword drives /change-password and ChangePasswordFragment.
- Nav config synced: web dashboardUi.js ↔ mobile DashboardUiCopy.kt (including Settings tab).
- Reusable UI: ConfirmDialog (web), MaterialAlertDialogBuilder (mobile), shared password form on web.

## 4. Screenshots of Working Features

Capture these views while the backend, web, and/or Android app are running. Placeholder frames below mark the required screens for the submission package.

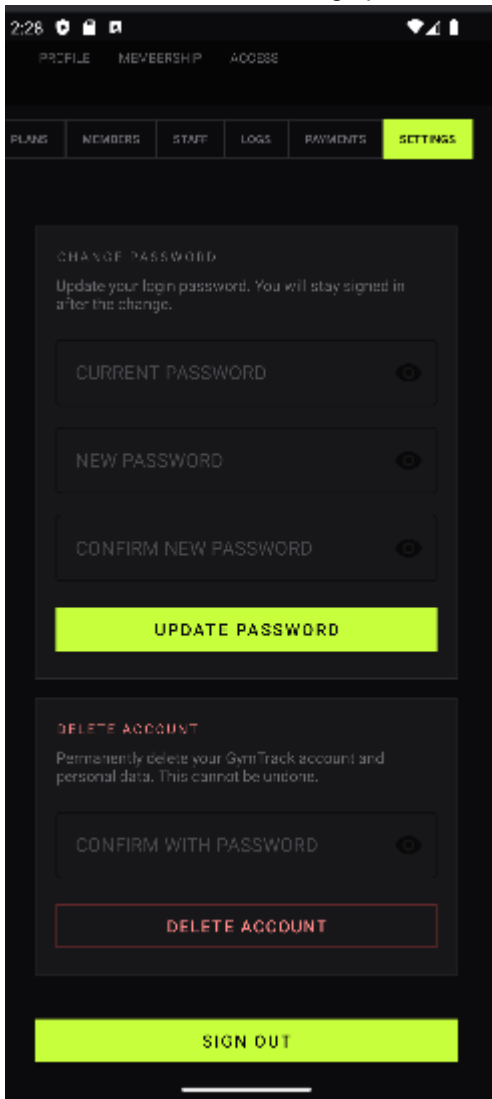
### Screenshot A — Web Settings tab

Dashboard → Settings: Change Password form and Delete Account section (all roles).



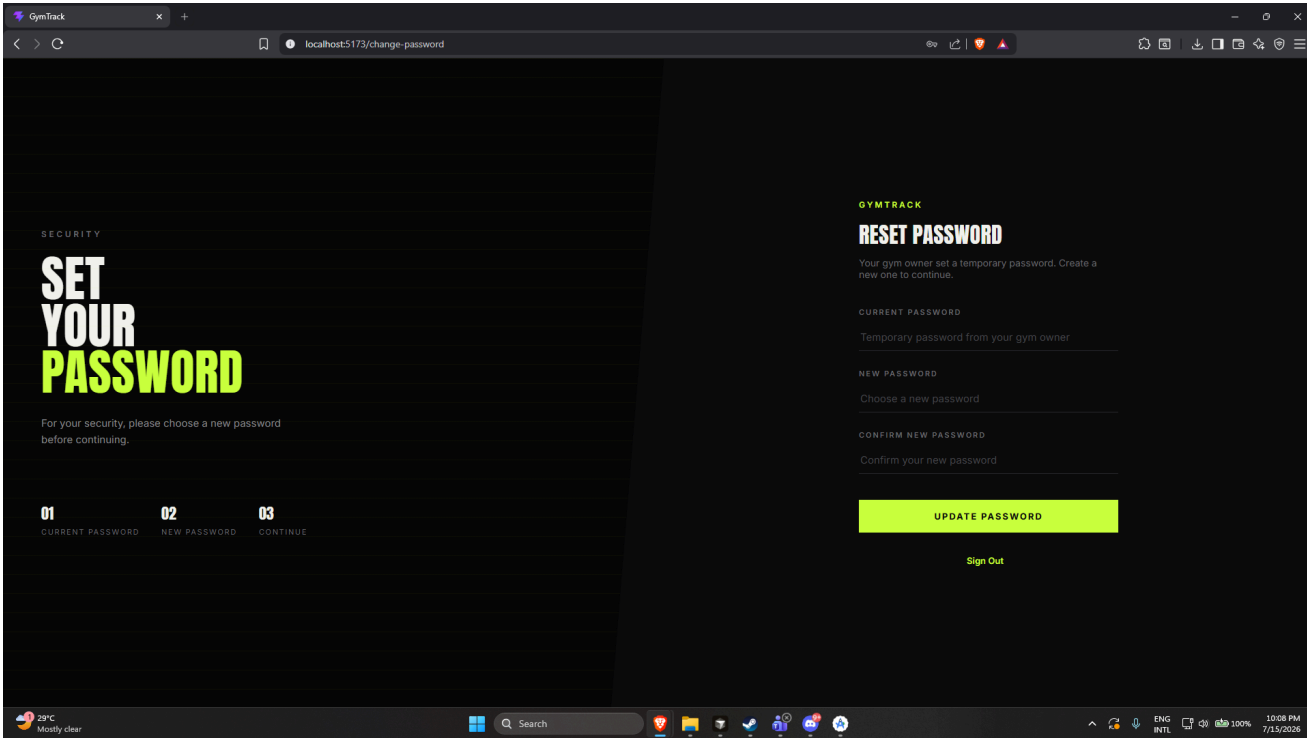
## Screenshot B — Mobile Settings tab

Android dashboard Settings panel with password fields and Delete Account action.



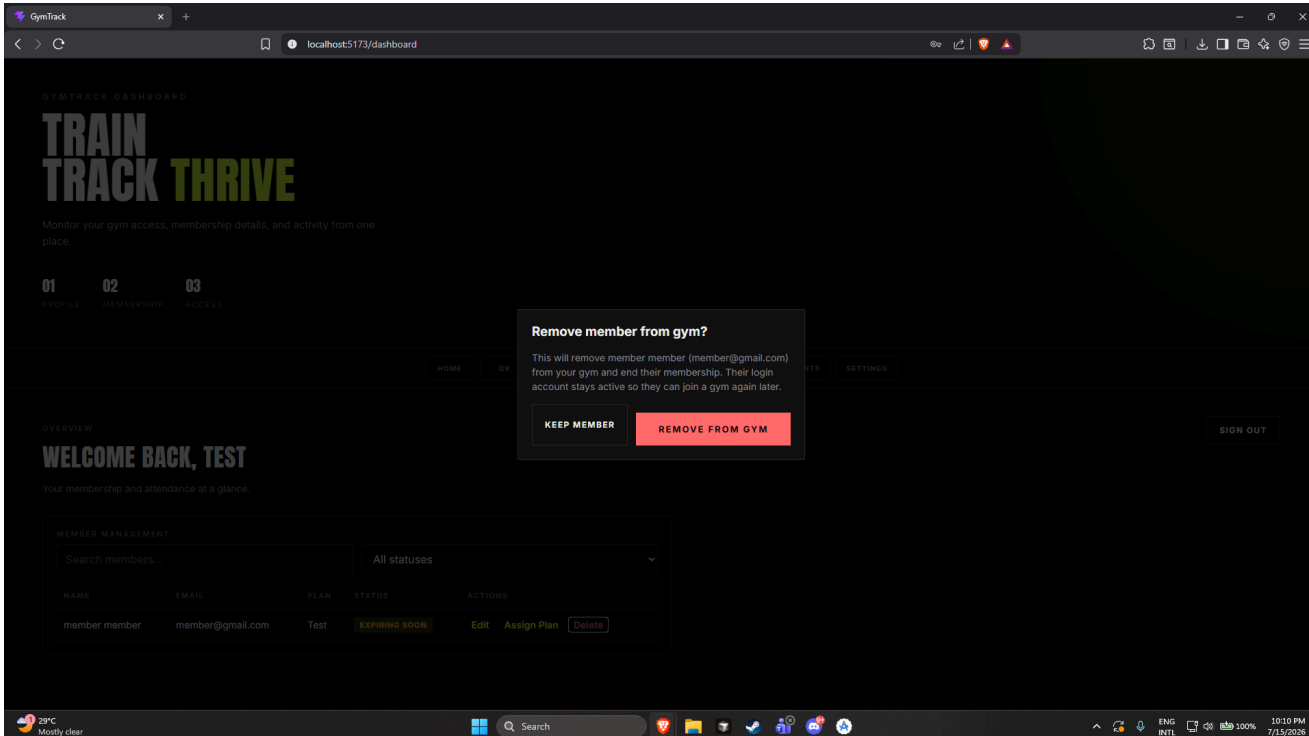
## Screenshot C — Staff first-login password reset

Forced reset password screen after staff signs in with a temporary password.



## Screenshot D — Admin remove member confirmation

Confirm dialog stating the member is removed from the gym; account is not deleted from the database.



## Screenshot E — Member QR / attendance (integrated core flow)

Member QR code or staff scan flow confirming attendance integration still works.

## 5. End-to-End Workflow Test Results

| Test #           | Workflow                                                              | Steps                                                                                                                                          | Expected                                                                                                      | Result                        |
|------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------|
| E2E-1            | Staff first login →<br>change password →<br>dashboard                 | Admin creates staff → staff logs in →<br>forced password page → sets new<br>password → dashboard loads → can<br>log in again with new password | mustChangePassword<br>cleared; JWT session<br>continues; no login loop                                        | PASSED                        |
| E2E-2            | Admin removes<br>member from gym<br>(soft unenroll)                   | Admin opens Members → Remove →<br>confirm → list refreshes → check<br>users table for same email                                               | Member disappears from<br>gym list; users row remains<br>with gym_id NULL;<br>memberships expired             | PASSED                        |
| E2E-3            | Settings: change<br>password then delete<br>account                   | Any role → Settings → update<br>password → sign in again → Settings<br>→ enter password → Delete Account<br>→ confirm                          | Password updated<br>successfully; after delete,<br>login fails and related<br>personal records are<br>removed | PASSED                        |
| E2E-4<br>(bonus) | Member PayMongo /<br>mock pay →<br>membership active →<br>QR check-in | Member enrolls via gym QR →<br>chooses plan → completes checkout<br>(mock/test) → shows member QR →<br>staff scans                             | Membership ACTIVE;<br>attendance log created                                                                  | PASSED<br>(mock/test<br>mode) |

## 6. Integration Issues Encountered and Solutions

| Issue                                                       | Cause                                                                                                             | Solution                                                                                                                                     |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Admin "delete member" wiped the user from the database      | DELETE /api/members/{id} called a hard-delete helper; also old Spring Boot process kept serving previous bytecode | Changed MemberService to unenroll (clear gym, expire memberships). Restart backend after code changes. Soft-remove confirmed in users table. |
| Login / Create Account buttons stayed disabled while typing | Enabled state only updated when ViewModel uiState emitted, not on text change                                     | Added local updateButtonState() on doAfterTextChanged for login and register (mobile).                                                       |

|                                                                          |                                                             |                                                                                                                                             |
|--------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Staff could not log in smoothly after first password change              | mustChangePassword / session routing edge cases after reset | Standardized AuthResponse flag handling and post-change navigation to dashboard; regression fix commit b5d441d.                             |
| Deploy security risk (open routes / weak secrets)                        | Role checks incomplete; prod secrets not validated          | Added method security, HTTP matchers, application-prod profile validator, webhook signature requirement.                                    |
| Web & mobile Settings needed identical behavior without theme complexity | Risk of duplicating password/delete logic per platform      | Reusable ChangePasswordForm + SettingsPanel (web); SettingsRepository reuses ChangePasswordRepository (mobile); shared /api/auth endpoints. |

## 7. Commit History Table

Selected commits mapped to integration work. Full history is on GitHub main branch.

| Feature / Change                                               | GitHub Commit Link                                                                                                      |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| User Settings: change password & delete account (web + mobile) | <a href="https://github.com/louisflim/gymtrack/commit/180a35b">https://github.com/louisflim/gymtrack/commit/180a35b</a> |
| Admin remove member from gym (soft unenroll, confirm dialog)   | <a href="https://github.com/louisflim/gymtrack/commit/315b9a3">https://github.com/louisflim/gymtrack/commit/315b9a3</a> |
| Fix staff login after forced first-login password change       | <a href="https://github.com/louisflim/gymtrack/commit/b5d441d">https://github.com/louisflim/gymtrack/commit/b5d441d</a> |
| Security: API role hardening and production secrets            | <a href="https://github.com/louisflim/gymtrack/commit/2ff61af">https://github.com/louisflim/gymtrack/commit/2ff61af</a> |
| Staff first-login password change (mustChangePassword)         | <a href="https://github.com/louisflim/gymtrack/commit/bbcb15c">https://github.com/louisflim/gymtrack/commit/bbcb15c</a> |
| User-friendly error messages (web, mobile, backend)            | <a href="https://github.com/louisflim/gymtrack/commit/1f77a7e">https://github.com/louisflim/gymtrack/commit/1f77a7e</a> |
| Fix mobile register Create Account button enable state         | <a href="https://github.com/louisflim/gymtrack/commit/5ffd37">https://github.com/louisflim/gymtrack/commit/5ffd37</a>   |
| Admin member/staff delete option & PayMongo fixes              | <a href="https://github.com/louisflim/gymtrack/commit/d3f259d">https://github.com/louisflim/gymtrack/commit/d3f259d</a> |
| QR code bugfix and UI updates                                  | <a href="https://github.com/louisflim/gymtrack/commit/be3010b">https://github.com/louisflim/gymtrack/commit/be3010b</a> |
| Fix Sign In button not enabling on login                       | <a href="https://github.com/louisflim/gymtrack/commit/6f58dfb">https://github.com/louisflim/gymtrack/commit/6f58dfb</a> |
| Gym entity bug fixes                                           | <a href="https://github.com/louisflim/gymtrack/commit/44e7acb">https://github.com/louisflim/gymtrack/commit/44e7acb</a> |
| Phase 2 documentation PDF                                      | <a href="https://github.com/louisflim/gymtrack/commit/4088f3b">https://github.com/louisflim/gymtrack/commit/4088f3b</a> |
| Vertical slice: staff accounts                                 | <a href="https://github.com/louisflim/gymtrack/commit/65f2c36">https://github.com/louisflim/gymtrack/commit/65f2c36</a> |
| Vertical slice: admin dashboard KPIs                           | <a href="https://github.com/louisflim/gymtrack/commit/61cd3ea">https://github.com/louisflim/gymtrack/commit/61cd3ea</a> |
| Vertical slice: QR attendance check-in/out                     | <a href="https://github.com/louisflim/gymtrack/commit/7740660">https://github.com/louisflim/gymtrack/commit/7740660</a> |
| Vertical slice: PayMongo checkout payments                     | <a href="https://github.com/louisflim/gymtrack/commit/f7f26f7">https://github.com/louisflim/gymtrack/commit/f7f26f7</a> |
| Vertical slice: membership status                              | <a href="https://github.com/louisflim/gymtrack/commit/8ac7024">https://github.com/louisflim/gymtrack/commit/8ac7024</a> |
| Vertical slice: member management                              | <a href="https://github.com/louisflim/gymtrack/commit/887c770">https://github.com/louisflim/gymtrack/commit/887c770</a> |
| Vertical slice: plan management                                | <a href="https://github.com/louisflim/gymtrack/commit/672bfbf">https://github.com/louisflim/gymtrack/commit/672bfbf</a> |
| Vertical slice: admin staff creation                           | <a href="https://github.com/louisflim/gymtrack/commit/3407f40">https://github.com/louisflim/gymtrack/commit/3407f40</a> |
| Vertical slice: user login                                     | <a href="https://github.com/louisflim/gymtrack/commit/544bab8">https://github.com/louisflim/gymtrack/commit/544bab8</a> |
| Vertical slice: user registration                              | <a href="https://github.com/louisflim/gymtrack/commit/e87ed8d">https://github.com/louisflim/gymtrack/commit/e87ed8d</a> |
| Layout and UI updates                                          | <a href="https://github.com/louisflim/gymtrack/commit/0e3151b">https://github.com/louisflim/gymtrack/commit/0e3151b</a> |
| Mobile login and register                                      | <a href="https://github.com/louisflim/gymtrack/commit/d47a6ce">https://github.com/louisflim/gymtrack/commit/d47a6ce</a> |
| Initialize Android mobile app                                  | <a href="https://github.com/louisflim/gymtrack/commit/38ec787">https://github.com/louisflim/gymtrack/commit/38ec787</a> |
| Admin/member dashboard                                         | <a href="https://github.com/louisflim/gymtrack/commit/6b9b568">https://github.com/louisflim/gymtrack/commit/6b9b568</a> |
| Project README                                                 | <a href="https://github.com/louisflim/gymtrack/commit/2b11064">https://github.com/louisflim/gymtrack/commit/2b11064</a> |
| Initial Commit                                                 | <a href="https://github.com/louisflim/gymtrack/commit/b9449f6">https://github.com/louisflim/gymtrack/commit/b9449f6</a> |